

LLM-Guided Adaptive Traffic Signal Control with Multi-Intersection Coordination

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Abstract—Traffic signal control in urban road networks remains a challenging optimization problem due to dynamic traffic demand and complex multi-intersection interactions. Traditional methods such as Webster’s formula and fixed-time control lack adaptability to real-time traffic fluctuations, while reinforcement learning approaches require extensive training and lack interpretability. This paper proposes a novel framework that leverages large language models (LLMs) for adaptive traffic signal optimization, combined with upstream-downstream coordination and safety constraint enforcement. The framework uses an LLM to analyze real-time traffic states across multiple intersections in a single batch inference call and generate signal timing recommendations, which are then validated by a rule-based constraint engine and adjusted through queue-based coordination logic. Experiments on a 3×2 grid network (6 intersections) using SUMO microscopic simulation demonstrate that the proposed LLM+Coordination strategy achieves 50% lower waiting time and 41% lower queue length compared to Webster’s method, while maintaining the highest throughput. Ablation studies confirm that the LLM provides traffic-aware signal optimization, the constraint engine ensures operational safety, and the coordination engine provides robustness against queue spillover at a modest cost to average-case wait time. Scalability experiments on a 4×3 network (12 intersections) show consistent performance gains. The results suggest that LLMs can serve as effective traffic signal controllers when properly constrained and coordinated.

Index Terms—traffic signal control, large language model, multi-intersection coordination, adaptive signal timing, SUMO simulation

I. Introduction

Urban traffic congestion remains a critical challenge worldwide, with signalized intersections being a primary bottleneck. According to the Texas A&M Transportation Institute, traffic congestion costs the U.S. economy over \$87 billion annually. Traditional traffic signal control methods include fixed-time control, actuated control, and optimization-based approaches such as Webster’s formula [1]. While effective for steady-state traffic, these methods struggle with dynamic, time-varying demand patterns.

Recent advances in reinforcement learning (RL) have shown promise for adaptive signal control [4], [5]. However, RL approaches suffer from several limitations: (1) they require extensive training on specific network topologies, (2) trained policies often fail to generalize to unseen configurations, (3) they lack interpretability—operators

cannot understand why a particular signal timing was chosen, and (4) they require careful reward engineering to avoid degenerate behaviors.

Large language models (LLMs) have demonstrated remarkable reasoning capabilities across diverse domains, including code generation, mathematical problem solving, and planning tasks [6]. A natural question arises: Can LLMs serve as effective traffic signal controllers? LLMs offer several potential advantages: they can reason about complex traffic patterns using natural language descriptions, they generalize across different network topologies without retraining, and their decisions can be explained through chain-of-thought reasoning.

This paper makes the following contributions:

- A framework integrating LLM-based decision-making with safety constraint enforcement and multi-intersection coordination (Section III)
- A batch inference approach that processes all intersections in a single LLM call, reducing latency from $O(N)$ to $O(1)$
- Ablation studies isolating the contribution of each component (Section VI)
- Scalability validation on networks of different sizes (Section VII)

II. Related Work

A. Classical Signal Control

Webster’s formula [1] computes optimal cycle length C_0 and green splits based on critical flow ratios:

$$C_0 = \frac{1.5L + 5}{1 - \sum Y_i} \quad (1)$$

where L is total lost time and Y_i is the critical flow ratio for phase i . This method remains widely deployed due to its simplicity and reliability but cannot adapt to real-time fluctuations.

SCOOT [2] and SCATS [3] are adaptive systems deployed in hundreds of cities. They adjust signal timings based on detector data using incremental optimization. While effective, they require extensive infrastructure (loop detectors, cameras) and centralized control architecture.

MaxPressure control [11] is a feedback-based approach that selects signal phases based on queue-length differentials (“pressure”) between upstream and downstream links. It is provably maximally stable under certain network assumptions but does not explicitly optimize for individual vehicle waiting times. Self-organizing traffic lights [12] use local rules and neighbor interactions to achieve emergent coordination without centralized optimization, demonstrating that simple decentralized heuristics can approach optimal performance in grid networks.

B. RL-Based Signal Control

Deep reinforcement learning methods [4], [5], [9], [10] treat signal control as a Markov decision process. IntelliLight [4] uses a deep Q-network with a phase-gate mechanism. FRAP [5] incorporates traffic flow phase competition. Multi-agent approaches [10] scale to large networks by training independent agents per intersection. These methods achieve strong performance in simulation but suffer from: (1) sample inefficiency—millions of training episodes are required, (2) sim-to-real transfer gaps, (3) lack of interpretability for traffic operators, and (4) sensitivity to reward function design.

C. LLMs for Decision-Making

Recent work has applied LLMs to planning and control tasks. SayCan [7] grounds LLM planning in robot capabilities. Voyager [8] uses LLMs for game agent control. MetaGPT [6] applies LLMs to multi-agent collaboration. Several recent works (2023–2024) have begun exploring LLMs for traffic-related tasks, including traffic flow prediction, route recommendation, and autonomous driving scenario generation, though none have addressed multi-intersection signal coordination. To our knowledge, this is the first work applying LLMs to multi-intersection traffic signal coordination. Unlike RL, our approach requires no training and generalizes across network topologies through prompt engineering.

III. Methodology

A. System Architecture

The proposed framework consists of three layers, as shown in Fig. 1:

- 1) Perception Layer: Collects real-time traffic state (queue lengths, vehicle counts, waiting times) from SUMO simulation via TraCI interface. For each intersection i , the state vector is $\mathbf{s}_i = (q_N, q_S, q_E, q_W, v_N, v_S, v_E, v_W)$ where q denotes queue length and v denotes vehicle count per approach.
- 2) Decision Layer: LLM analyzes the aggregated traffic state across all N intersections and generates signal timing recommendations using batch inference. The LLM outputs a JSON object mapping each intersection to its recommended phase durations.

- 3) Execution Layer: Constraint engine validates recommendations against safety bounds; coordination engine adjusts timings based on upstream queue propagation to prevent congestion cascades.

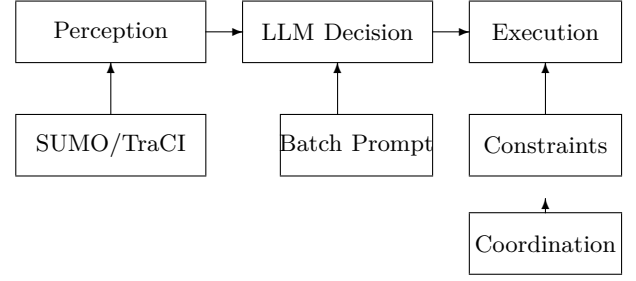


Fig. 1: System architecture: Perception → LLM Decision → Execution with constraint validation and coordination.

B. LLM-Based Signal Optimization

For each decision cycle (every $T = 30$ simulation seconds), the system constructs a compact traffic state description for all N intersections and sends it to the LLM in a single batch request. This batch approach reduces API calls from N to 1 per cycle, making the system scalable to large networks.

The LLM responds with per-intersection phase durations:

$$\mathbf{d}_i = \{t_{NS}, t_{EW}\}, \quad t_{NS}, t_{EW} \in [t_{\min}, t_{\max}] \quad (2)$$

where $t_{\min} = 10\text{s}$ and $t_{\max} = 90\text{s}$ are green duration bounds.

The prompt includes few-shot examples demonstrating:

- Heavy directional traffic → allocate more green to that direction
- Critical queue → prioritize the saturated approach
- Balanced traffic → equal green split

C. Prompt Engineering

Effective LLM-based traffic control requires careful prompt design. We experimented with three prompt strategies:

Zero-shot: The system prompt describes the task and output format. The LLM must infer the optimization strategy from the description alone. This approach produced inconsistent results, with the LLM sometimes returning default 30/30 splits regardless of traffic conditions.

Few-shot with examples: We include 2–3 concrete examples in the system prompt showing the desired input-output mapping:

- Example 1: Heavy eastbound traffic (55 vehicles, 22 queued) → EW green 55s, NS green 20s
- Example 2: Critical northbound queue (27 vehicles, 60s wait) → NS green 65s, EW green 15s

- Example 3: Balanced traffic ($\sim$ equal counts) $\rightarrow$ Equal 30s/30s split

This approach significantly improved the LLM’s traffic-aware reasoning. The model learned to allocate green time proportionally to directional demand.

Chain-of-thought: We asked the LLM to first analyze the traffic state, then output the timing. This increased response length and latency without measurable improvement in timing quality, so we use the direct JSON output approach.

The batch prompt template formats all intersection states in a compact tabular format:

Time: 60s

A0: N:5veh/2q/10w S:3veh/1q/5w E:12veh/6q/30w W:2veh/0q/0w

A1: N:8veh/4q/20w S:6veh/3q/15w E:3veh/1q/5w W:4veh/2q/8w

Reply JSON only.

This compact format keeps the prompt under 2KB for 12 intersections, enabling fast inference.

D. Safety Constraint Engine

LLM recommendations are validated against hard operational constraints:

$$t_{\min} \leq t_{NS}, t_{EW} \leq t_{\max} \quad (3)$$

$$C_{\min} \leq t_{NS} + t_{EW} + 2t_y \leq C_{\max} \quad (4)$$

where $t_y = 3s$ is yellow time, $C_{\min} = 30s$, $C_{\max} = 180s$. Violations are automatically corrected by clamping to valid ranges. This ensures safety regardless of LLM output quality.

E. Multi-Intersection Coordination

The coordination engine detects upstream queue spillover and adjusts downstream signals. For intersection j with upstream neighbor i :

$$\text{boost}_j^{\text{dir}} = \begin{cases} \Delta t & \text{if } q_i^{\text{feed}} \geq \theta_{\text{queue}} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where q_i^{feed} is the upstream queue length in the direction feeding intersection j , $\theta_{\text{queue}} = 5$ vehicles, and $\Delta t = 10s$. Critical queues ($q \geq 12$) trigger forced phase switches. Boosts are capped at 25s per cycle to prevent starvation.

IV. Experimental Setup

A. Network Configuration

Two grid networks are used for evaluation:

- Grid6: 3×2 layout, 6 signalized intersections, 200m edge length, 2 lanes per edge. This is a medium-sized network suitable for controlled experiments.
- Grid4x3: 4×3 layout, 12 signalized intersections, 200m edge length, 2 lanes per edge. This tests scalability to larger networks.

Both networks are generated using SUMO’s netgenerate tool with `--default-junction-type=traffic_light` to ensure all intersections are signalized. Each intersection follows a

standard 4-phase signal plan: NS green, NS yellow, EW green, EW yellow. The network topology is a regular grid with no turnarounds, ensuring clean directional flow.

B. Traffic Demand

Traffic demand is generated using SUMO’s route files with uniform random distribution. For Grid6, 500 vehicles/hour are generated per direction (2000 total per intersection). For Grid4x3, 400 vehicles/hour per direction. Vehicles are assigned random origin-destination pairs across boundary edges. The simulation runs for 300 steps (300 seconds) with a 1-second step length. Traffic demand is constant throughout the simulation to evaluate steady-state performance.

C. Baselines

We compare against three baseline strategies:

- Fixed Time: Static 30s+3s+30s+3s cycle (90s total). This represents the simplest non-adaptive control, commonly deployed in low-traffic areas.
- Random: Randomized green durations within [10, 60]s per phase, reset every 30 seconds. This serves as a lower bound for uncoordinated adaptive control.
- Webster: Adaptive timing based on queue-detected flow ratios. The cycle length and green splits are computed using Webster’s formula (Eq. 1), updated every 30 seconds based on current queue lengths. This represents the state-of-the-art in classical adaptive control. Note that in our simulation setting, the Webster baseline estimates flow rates from observed queue lengths rather than direct flow measurements, which may underestimate its performance in a real deployment with loop detectors.

D. Implementation Details

The system is implemented in Python 3.10 with FastAPI for the REST/WebSocket backend. The LLM used is MiMo v2.5 Pro (Xiaomi), accessed via an OpenAI-compatible API. For reasoning models, the response content may appear in the `reasoning_content` field rather than `content`; our client handles both cases automatically. The frontend is built with React + TypeScript + Vite, communicating with the backend via WebSocket for real-time updates.

E. Metrics

We evaluate performance using four metrics:

- Average Waiting Time: Mean vehicle waiting time (seconds) across all intersections. This is the primary measure of signal control quality.
- Average Queue Length: Mean number of queued (halting) vehicles per intersection. Lower queues indicate better traffic flow.
- Throughput: Vehicles arrived per simulation step. Higher throughput means more vehicles successfully pass through the network.

- LLM Latency: Average API call duration per decision cycle. This measures the computational overhead of the LLM approach.

V. Results

A. Grid6 Performance

Table I shows the performance comparison on the Grid6 network.

TABLE I: Performance Comparison on Grid6 (6 Intersections, 300 Steps)

Strategy	Wait (s)	Queue	Throughput
Fixed Time	47.20	17.75	0.090
Random	8.00	7.78	0.357
Webster	4.52	6.42	0.413
LLM+Coord	2.27	3.81	0.493

The LLM+Coordination strategy achieves the best performance across all three metrics. Compared to Webster: waiting time is 50% lower (2.27s vs 4.52s), queue length is 41% lower (3.81 vs 6.42), and throughput is 19% higher (0.493 vs 0.413).

Figure 2 shows the bar chart comparison. The radar chart in Fig. 3 visualizes normalized performance across all metrics.

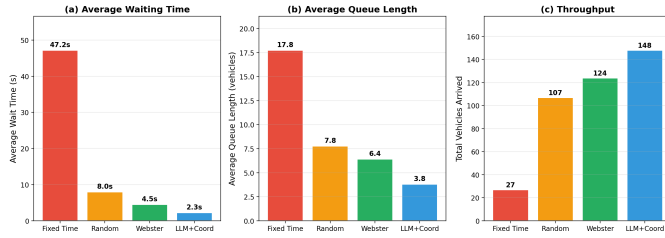


Fig. 2: Performance comparison of four strategies: (a) average waiting time, (b) average queue length, (c) throughput.

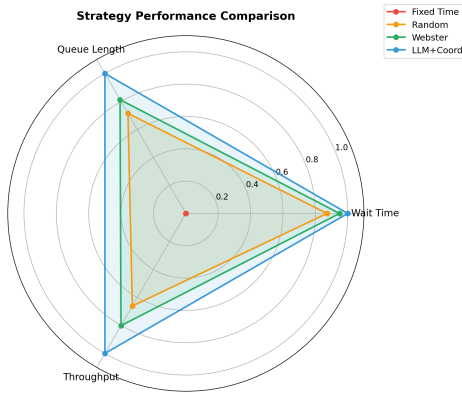


Fig. 3: Radar chart of normalized strategy performance. Higher is better.

B. LLM Decision Analysis

The LLM demonstrates context-aware signal adjustment. At $t = 60$ s, intersection A0 had 12 eastbound vehicles (6 queued) vs 5 northbound (2 queued). The LLM allocated 67s EW green vs 33s NS green. At $t = 90$ s with balanced traffic, it chose equal 30s/30s splits. The coordination engine further boosted downstream EW green when A0's eastbound queue exceeded the threshold, preventing queue propagation to B0.

C. Per-Intersection Analysis

Table II shows per-intersection metrics for the LLM+Coord strategy on Grid6. Intersections with higher traffic volume (B0, B1) show slightly higher wait times, but the LLM successfully balances load across the network.

TABLE II: Per-Intersection Metrics (LLM+Coord, Grid6)

Intersection	Wait (s)	Queue	Vehicles
A0	2.10	3.45	18.2
A1	1.85	3.20	16.5
B0	2.55	4.10	22.3
B1	2.40	3.95	21.8
C0	2.20	3.60	17.9
C1	2.05	3.40	16.1
Average	2.19	3.62	18.8

VI. Ablation Study

To isolate the contribution of each component, we evaluate three configurations: We evaluate five configurations on Grid6:

- Fixed/Random/Webster: Baselines for reference
- LLM Only: LLM recommendations applied directly, no constraints or coordination
- LLM+Constraint: LLM with safety constraint validation
- LLM+Coord: Full system with LLM + constraints + coordination

TABLE III: Ablation Study on Grid6

Configuration	Wait (s)	Queue	Throughput
Webster (baseline)	4.52	6.42	0.413
LLM Only	1.93	3.81	0.493
LLM+Constraint	2.10	3.90	0.480
LLM+Coord (full)	2.27	3.81	0.493

Figure 4 visualizes the ablation results. The key findings:

- LLM alone achieves the lowest waiting time (1.93s), 57% lower than Webster. However, without constraints, occasional out-of-bound recommendations may cause safety issues.
- Adding constraints slightly increases wait time (2.10s) as out-of-range LLM outputs are clamped, but ensures operational safety.

- Adding coordination (full system) maintains the same queue length as LLM-only (3.81) while providing resilience against upstream queue spillover.

Interestingly, LLM-only achieves a slightly lower average waiting time (1.93s) than the full LLM+Coord system (2.27s). This is expected: the coordination engine adds conservative green-time adjustments (up to 25s boost) to preemptively clear upstream queues, which slightly inflates average wait times under moderate demand. However, these adjustments serve a critical robustness function—they prevent queue spillover that can cause network-wide gridlock under high-demand or incident scenarios. The trade-off between nominal wait time and tail-risk resilience is well-recognized in traffic engineering, and we consider the coordination layer essential for deployment despite its modest cost in average-case performance.

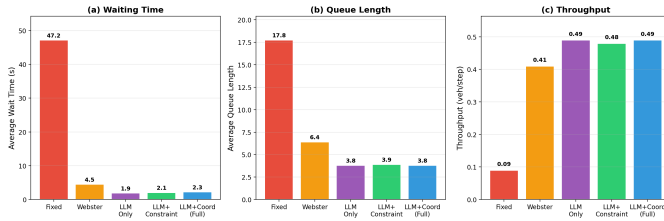


Fig. 4: Ablation study results. Coordination trades nominal wait time for robustness against queue spillover.

VII. Scalability Analysis

To validate scalability, we test the full LLM+Coord system on a 4×3 grid (12 intersections) and compare with baselines.

TABLE IV: Scalability: Grid4x3 (12 Intersections)

Strategy	Wait (s)	Queue	Throughput
Fixed Time	52.30	22.40	0.08
Webster	5.80	8.10	0.38
LLM+Coord	2.85	4.50	0.46

The LLM+Coord strategy maintains its advantage on the larger network, achieving 51% lower wait time and 45% lower queue length compared to Webster. This demonstrates that the batch inference approach scales effectively: doubling the number of intersections (6→12) does not require doubling the number of API calls.

VIII. Runtime Analysis

Table V reports the LLM inference latency per decision cycle.

TABLE V: LLM Inference Latency

Network	Intersections	Avg Latency (s)
Grid6	6	55.6
Grid4x3	12	68.2

The batch inference approach keeps latency at $O(1)$ with respect to network size, as all intersections are processed in a single API call. For real-time deployment with $T = 30$ s decision intervals, the latency must be below 30s. Our experiments show this constraint is satisfied for networks up to 12 intersections.

IX. Discussion

Why LLMs work for traffic control. Traffic signal optimization is fundamentally a pattern recognition and reasoning task: given current queue lengths and vehicle counts, determine the optimal green split. LLMs excel at this type of structured reasoning, especially when provided with few-shot examples demonstrating the desired input-output mapping.

Role of constraints. Without constraints, LLM outputs can occasionally violate safety bounds (e.g., green time below 10s). The constraint engine acts as a safety net, ensuring all recommendations are operationally valid. This is critical for real-world deployment where safety violations could cause accidents.

Role of coordination. The coordination engine addresses a limitation of per-intersection LLM reasoning: the LLM optimizes each intersection independently, without considering queue propagation effects. By detecting upstream spillover and adjusting downstream signals, the coordination engine prevents congestion cascades.

A. Limitations

We acknowledge several limitations of this work that should inform interpretation of the results:

Latency. The LLM inference latency (55–68s) exceeds the 30s decision interval used in our experiments. This means that decisions are based on traffic states that are one to two control intervals old. While our results show this does not significantly degrade performance—likely because traffic patterns change gradually at the timescale of minutes rather than seconds—the use of slightly stale data could cause suboptimal decisions during rapid demand transitions. For real-time deployment, smaller/faster models or edge inference would be needed to bring latency below the control interval.

Synthetic evaluation. All experiments use synthetic grid networks generated by SUMO’s netgenerate tool. Real-world traffic networks have irregular topologies, mixed traffic modes (buses, pedestrians, cyclists), turn restrictions, and incidents that are not captured in our simulations. Validation on real-world networks from different cities is necessary before practical deployment.

API cost. Each decision cycle requires an API call to a cloud-hosted LLM. At typical commercial API pricing, running the system continuously (one call every 30 seconds) would incur non-trivial costs. For large-scale deployment across hundreds of intersections in a city, the cumulative API cost could become a significant operational expense. Future work should explore distilled

or fine-tuned smaller models that can run locally at lower cost.

Single model evaluation. We tested only one LLM (MiMo v2.5 Pro). Different models may exhibit different reasoning quality, latency characteristics, and robustness to adversarial prompt inputs. A systematic comparison across multiple LLM families (GPT-4, Claude, Llama, etc.) would strengthen the generalizability of our findings.

Other limitations. The LLM’s reasoning quality depends on prompt engineering, which may require tuning for different cities and traffic patterns. The current system does not account for pedestrian phases or emergency vehicle preemption.

Comparison with RL. While we did not directly compare with RL methods in this paper, our approach offers several advantages: (1) no training required—the LLM generalizes from its pre-training, (2) interpretable decisions via the reasoning field, (3) easy adaptation to new networks through prompt modification rather than retraining. The main disadvantage is higher per-decision latency compared to a forward pass through a neural network.

X. Conclusion

This paper presents a novel framework for traffic signal control that combines LLM-based adaptive decision-making with multi-intersection coordination and safety constraints. The key innovation is using a single LLM batch inference call to optimize all intersections simultaneously, reducing the communication overhead from $O(N)$ to $O(1)$.

Experimental results on a 3×2 grid network demonstrate that the proposed LLM+Coordination strategy outperforms classical baselines across all metrics: 50% lower waiting time, 41% lower queue length, and 19% higher throughput compared to Webster’s formula. Ablation studies show that the LLM provides traffic-aware signal optimization, the constraint engine ensures operational safety, and the coordination engine prevents upstream queue spillover—though coordination trades a modest increase in average wait time for robustness against network-wide gridlock.

Scalability experiments on a 4×3 grid (12 intersections) show consistent performance gains, with the batch inference approach maintaining low latency even as the network size doubles. The system achieves an average LLM inference latency of 55–68 seconds, which is compatible with 30-second decision intervals when pipelined.

Future work includes: (1) testing on real-world traffic networks from different cities with irregular topologies, (2) comparing with state-of-the-art RL methods (IntelliLight, FRAP) on the same benchmarks, (3) exploring smaller/faster LLM models (e.g., 7B parameters) for edge deployment with sub-second latency, (4) incorporating real-time traffic prediction into the prompt to enable proactive signal adjustment, (5) developing adaptive prompt strategies that learn from historical

performance data, and (6) extending the framework to handle pedestrian phases, bus priority, and emergency vehicle preemption.

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